

De Montfort University

Beng (Hons) Mechanical Engineering – Year 3

Objective: Design & Analyse the Suspension & Steering Components of the Future 2027 DMU
FSAE Race Car

Report: Suspension & Steering

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Abstract

The suspension & steering geometry must be designed for the formula student competition. Designing the suspension optimally requires collaboration with the upright and chassis designer to understand design limits and negotiation final designs. The aims and objectives of this design challenge are to gain experience in a team, against tight deadlines and compete against other teams applying engineering knowledge in real world conditions. The real-world limits are cost, manufacturing & time constraints. Suspension types must be researched and analysed to find the optimal suspension system for formula student racing. Using Susprog3d (a mathematical software calculating the optimal suspension node co-ordinates based on variables of a race car), the optimal suspension nodes must be obtained and used in 3D modelling software such as CREO to then build a 3D model of the suspension for FEA analysis. The model must then be improved and finalised to fit well with the chassis and upright geometry, where the chassis and upright designers must also modify their geometry to fit with the suspension geometry. The final design and findings will then be compiled and communicated to the Formula Student DMU race team for future use in competition.

Introduction

In the UK the Institution of Mechanical Engineers (IMechE) hosts the formula student competition. It is a competition among universities and takes place annually at Silverstone Circuit. University students compete to design and bring forth a race car under the formula student regulations to race against other universities. The competition is split into two different categories: Static Events (business/design) and Dynamic Events (on-track performance). Static Events cover the engineering design, cost/manufacturing and a business plan presentation (mock commercialisation). Dynamic Events cover a standing start acceleration test of 75 meters, a figure-eight timed circuit, a single lap, endurance and fuel efficiency during the race. Suspension and steering are crucial to making an efficient, fast and safe car in this race.

Aims

The aim of this project is to design:

- analyse and manufacture the suspension and steering components for the future DMU race team to race in formula student.
- Improve the cars performance in the formula student competition compared to last year.

Objectives

This project entails the following:

- Research of suspension and steering components.
- Which type of suspension is best for racing.
- What modifications or add-ons can be applied to improve performance.
- Details on how manufacturers provide these components, the cost, what materials to use and how the chassis/upright dimensions must be considered when designing the suspension.
- Using CREO, Ansys & Susprog3d to design the suspension components.
- How the components will be attached to the DMU team race car.
- Develop bill of materials and cost analysis for designed suspension/steering system.

Sustainability Goals [\[1\]](#)

1. SDG 4 (Quality Education): This project utilises advanced technical software such as CREO to design the components in 3D, Susprog3d to simulate the kinematics of suspension

according to key values and structural analysis using ANSYS FEA. This aligns with the goal of increasing technical skills of persons with the aim of being professional engineers.

2. SDG 9 (Industry, Innovation & Infrastructure): By using Finite Element Analysis (FEA) to optimise the component's weight and strength, this project promotes the efficient use of material and sustainable design innovation.
3. SDG 12 (Responsible Consumption and Production): Using 3D simulation software, less real-life physical prototypes will be wasted attempting to find a solution. Therefore, less materials are used in total to fulfil the same goal, and FEA means safety is not compromised.



Figure 1 United Nations Sustainable Development Goals

Deliverables:

- Common suspension systems, their advantages and disadvantages.
- Which ones are used for racing around a formula style track.
- Which one is best for DMU racing team car based on cost, manufacturability, effectiveness.
- Utilising Susprog3d to calculate optimal suspension nodes (points that will connect to car/wheel). Design and analysis using CREO and ANSYS.
- Planning a project that requires research analysis, design, collaboration & software skills.
- How to prioritise aspects of a car for a specific aim; should it be lightweight or heavy, complex or simple, cheap or expensive? And how to transfer that into a design.

Methodology

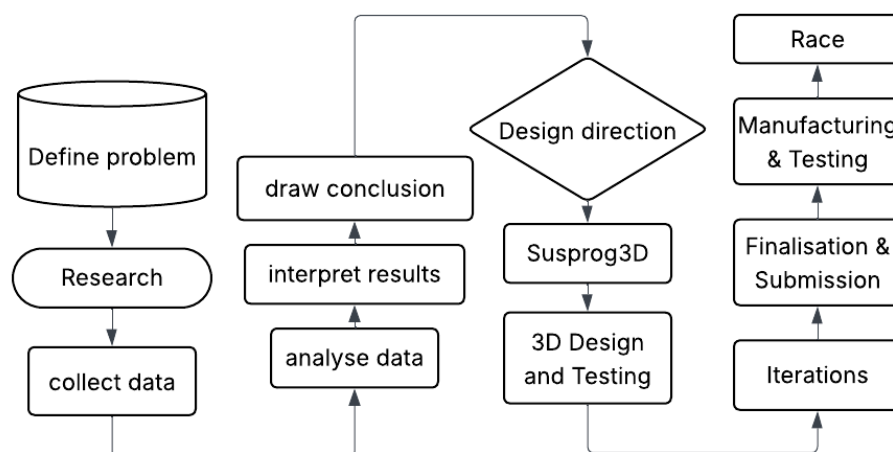


Figure 2 Methodology Steps Flow Chart

Research Gap

- How camber, caster, toe & king-pin inclination can be optimised for racing cars.

- Rules and regulations of formula student race that affects suspension and steering geometry.
- Chassis, Upright, Suspension/Steering dimension limitations.

Research

Title: On the Influence of Suspension Geometry on Steering Feedback [2]

Publisher: MDPI (Applied Sciences)

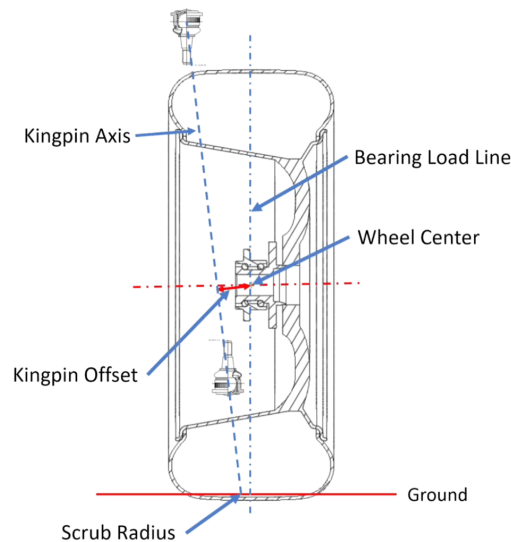


Figure 3 Kingpin Axis, Offset & Scrub Radius Diagram of a Wheel

Kingpin Offset (KO) is critical. Kingpin offset is the primary driver of torque steer and steering disturbance such as bumps or rocks on the road. The KO should be kept as small as possible (ideally near 10-15mm) to ensure the driver receives clean feedback from the tires rather than noise from the drivetrain. This must be negotiated with the upright designer due to possible dimensional limitations.

Scrub Radius (SR) can be lowered to reduce steering effort; however, some SR is necessary for the driver feel the limit of adhesion.

Improper geometry such as excessive KO or SR causes high force peaks when the wheels self-centre. In a formula student endurance event, this leads to driver mental fatigue and inconsistent lap times. The geometry must prioritise self-aligning torque to make the car intuitive to drive.

Title: Design & Analysis of Formula Student Race Car Suspension System [3]

Publisher: ScienceDirect (Procedia Engineering)

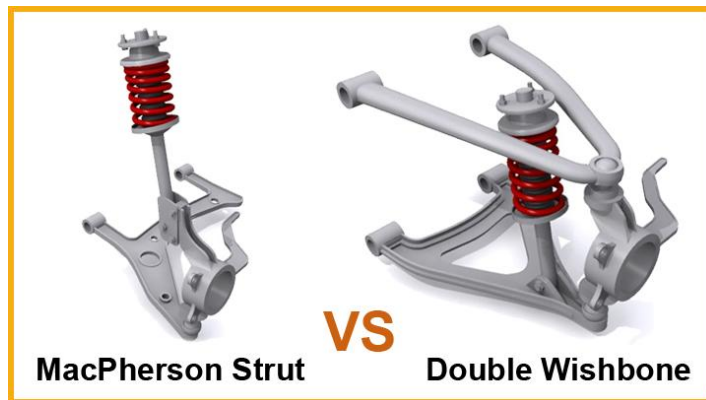


Figure 4 Comparison of Traditional Suspension Against Double Wishbone Suspension Geometry.

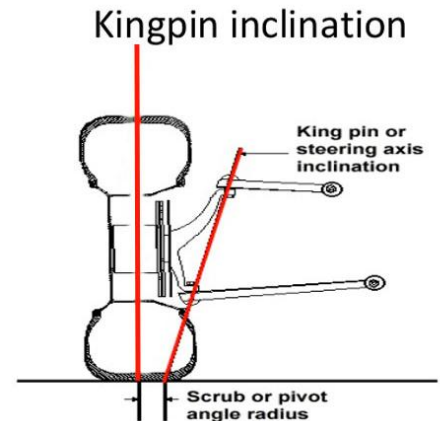


Figure 5 Kingpin Inclination and Scrub Radius

Unequal length non -parallel double wishbone suspension is superior in maintaining the tire contact patch during body roll.

The wishbone pick-up points on the chassis (points that connect to the chassis) must be reinforced to handle braking (longitudinal) and cornering (lateral) loads simultaneously without deflection.

The goal is high roll stiffness with minimal unsprung mass. FEA will be used to justify material choices.

Every gram saved on the uprights and wheels is worth double in performance.

Collaboration with the upright designer is required to ensure Kingpin Axis (the line between the upper and lower ball joints) allows for the desired Caster and Kingpin Inclination (KPI) while leaving room for the brake callipers and hubs.

Title: Race Car Design [4]

Publisher: Derek Seward

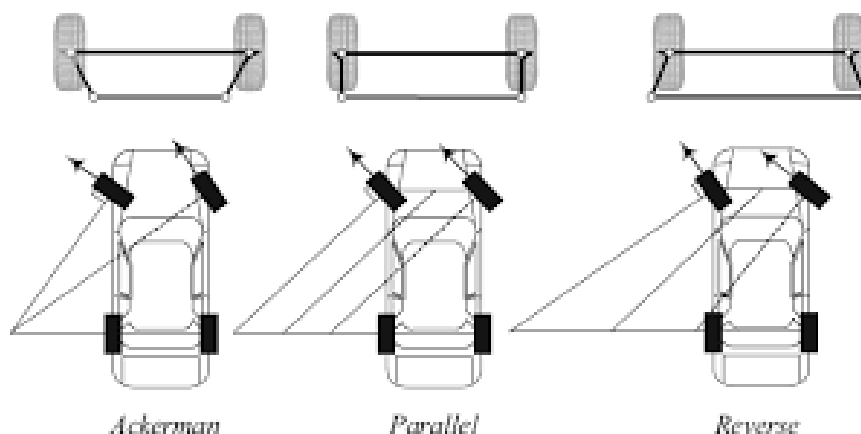


Figure 6 Ackermann vs Anti Ackermann (known as reverse) Geometry

Suspension is a compromise between ride (bumps) and roll (cornering). For FS, where tracks are usually flat but tight, roll control is more important.

Pro-Ackermann geometry optimal for tight hairpins typical of FS tracks. This will require the steering rack to be positioned specifically relative to the front axle line, which must be communicated early to the chassis designer to ensure space for the rack and steering column.

Title: Design Analysis & optimisation of Race Car Suspension [5]

Publisher: IJERT (Internation Journal of Engineering Research & Technology)



Figure 7 Tyre Wear Types from Camber/Caster/Toe & Geometries Diagram

Suspension geometry should be designed to provide negative camber gain during jounce. This counteracts the positive camber caused by body roll, keeping the tire flat on the ground.

The steering tie-rods must be perfectly aligned with the wishbones. Even small amounts like 1mm of error in the steering rack height can cause the wheels to steer themselves as the suspension moves up and down (bump steer), which is a “fail” in FS design judging.

Title: Formula Student Rules & Regulations 2026 [6]

Publisher: IMechE (Formula Student UK)

T4.3 Suspension Requirements: The car must have a functional suspension system with dampers on all four wheels and a minimum travel of 25mm (jounce & rebound)

T4.1 Steering Limits: The steering system must have “positive stops” to prevent the tires from hitting the chassis or suspension members at full lock. There must be zero slack in the steering.

T4.2 Wheelbase/Track: The wheelbase must be at least 1525mm. the smaller of the two tracks (front/rear) must be at least 75% of the larger track.

Rules require a specific tube thickness for the primary structure where suspension attaches. Car must maintain ground clearance of 30mm at any portion of the car except the wheels when a driver is in the

Outline Plan

- Table 1 Gantt Chart. Dark grey means should be completed by that week. Light grey means might take these extra weeks to complete. Light blue means it will be consistent until submission.

[illegible]

References

- [1] "United Nations 17 sustainable development goals". *Un.org*, 2015 <https://sdgs.un.org/goals> (March. 14, 2026).
- [2] "On the Influence of Suspension Geometry on Steering Feedback". *Uni of Brescia*, 2020 [On the Influence of Suspension Geometry on Steering Feedback | MDPI](#) (March. 15, 2026).
- [3] "Design & Analysis of Formula Student Race Suspension System". *Procedia Engineering*, 2016 <https://www.sciencedirect.com/science/article/pii/S1877705816302983> (March. 16, 2026).
- [4] "Race Car Design". *Derek Seward*, 2014 <https://books.google.co.uk/books?hl=en&lr=&id=s3atEAAQBAJ&oi=fnd&pg=PP1&dq=racecar+design&ots=XIO7UIFsDs&sig=FQtzxRBlkyO00IIGiTMhc-mNRak#v=onepage&q=racecar%20design&f=false> (March. 20, 2026).
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- [6] "Formula Student Rules & Regulations". IMechE, 2026 <https://www.imeche.org/docs/default-source/1-oscar/formula-student/2026/rules/fsuk-2026-rules---v1-09e21118e54216d0c8310ff0100d05193.pdf?sfvrsn=2> (March. 22, 2026).